Cubic Obstructions to Zero-Sector Reduction
and Arithmetic Defect Formulas

Research draft prepared for Keio

7 November 2026 (UTC)

Abstract

We give an exact negative answer to Question 6.1 of Cardon, Forgács, Piotrowski, Sorensen and White (2018). For $T_{\alpha,\beta}(z^k) = e^{-\alpha k^\beta} z^k$, the original proposed sufficiency of $\beta \geq 2$ fails already at $\beta = 3$ and $\alpha = \log(100/99)$: the image of $(z-1)^3$ has a strictly negative cubic discriminant. A general calculation gives an explicit third-finite-difference obstruction for diagonal exponential semigroups, and consequently a quadratic-generator rigidity theorem. Separately, an odd p-adic logarithm expansion proves the Wolstenholme criterion printed in Meštrović's 2015 Remark 1.1, including all small primes. We identify why this arithmetic conclusion is already an elementary consequence of earlier established congruences. Appendices prove the unchanged binomial power-sum assertion in his 2014 Conjecture 19, derive an exact local valuation refinement, and give a counterexample to the first assertion of Conjecture 11. All mathematical proofs are unconditional. Historical priority is not certified, and no result is claimed to satisfy all six novelty gates specified in the research brief. No proof of RH, BSD, abc, or infinitude of Wieferich or Wolstenholme primes is obtained.

Reading this draft. "Answer to a printed assertion" describes a proved mathematical conclusion. It does not, by itself, assert that the assertion was still open on the date of this draft or that the proof is historically new. Section 6 records these distinctions explicitly. The single `.tex` source contains all proofs, references, and an exact-arithmetic verification program; no external figures, bibliography files, or source fragments are required to compile it.

Keywords. Zero-sector reduction; multiplier sequences; cubic discriminant; finite differences; Wolstenholme primes; harmonic congruences; power sums.

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1 Targets, claims, and conventions

The principal target is the parameter criterion proposed in [1, Question 6.1]. Our counterexample uses the original diagonal operator and an admissible real polynomial, without replacing the target by a newly defined invariant. The quantitative semigroup problem studied afterward is auxiliary: its solution explains the counterexample but is not counted as an additional pre-existing named problem.

The arithmetic targets are identified by their original numbered locations. The following inventory distinguishes the conclusions from their research-status assessment.

Original location	Exact conclusion in this draft
Cardon et al. (2018), Question 6.1	Negative answer to the proposed "$\beta \geq 2$" sufficiency, with a rational cubic certificate.
Meštrović (2015), Remark 1.1	The displayed harmonic congruence is equivalent to the Wolstenholme condition. This follows from earlier arithmetic and is not claimed as a new method.
Meštrović (2014), Conjecture 19	The stated congruence holds for every required k and even n; an exact local valuation is also proved.
Meštrović (2014), Conjecture 11	The first nonexistence assertion is false at $n = 5$. The second assertion is not resolved here.

Throughout, v_p denotes the usual valuation on $\mathbb{Q}$, with $v_p(0) = +\infty$. A rational congruence $u \equiv v \pmod{p^r}$ means $v_p(u - v) \geq r$. When denominators are units at p, this is the usual congruence in $\mathbb{Z}_{(p)}$. We write $\Delta g(m) = g(m + 1) - g(m)$. Zeros are counted with multiplicity; repeated zeros are allowed in all polynomial tests below.

2 A negative answer to the power-exponent question of Cardon et al.

2.1 The original problem and its quantifiers

Following Cardon, Forgács, Piotrowski, Sorensen and White [1], set

$$S(\theta) = \{0\} \cup \{z \in \mathbb{C} \setminus \{0\} : |\arg z| \leq \theta\}, \qquad 0 \leq \theta < \frac{\pi}{2}.$$

These are sectors about the nonnegative real axis. Writing $Z(f)$ for the multiset of zeros of a nonzero polynomial f, the sector-reduction property is

$$\begin{aligned} &\text{for every } \theta \in (0, \pi/2) \text{ there exists } \gamma \in [0, \theta) \text{ such that} \\ &\quad Z(f) \subseteq S(\theta) \implies Z(Tf) \subseteq S(\gamma) \quad \text{for every nonzero } f \in \mathbb{R}[z]. \end{aligned} \tag{1}$$

All operators considered below have strictly positive diagonal entries, so Tf is nonzero whenever f is nonzero.

Question 6.1 of [1] asks whether, for a positive parameter α, the operator (writing β for the exponent called p in the original question)

$$T_{\alpha,\beta}[z^k] = \exp(-\alpha k^\beta) z^k, \qquad k = 0, 1, 2, \ldots, \tag{2}$$

is zero-sector reducing if and only if $\beta \geq 2$. The question appears on page 489 of the published paper and page 10 of arXiv:1802.02641v1. In particular, its proposed sufficient direction would make (1) hold for every $\alpha > 0$ and every $\beta \geq 2$. The next theorem disproves this direction with the original operator and the original sector convention.

Theorem 2.1 (An exact cubic counterexample)**.** *For every*

$$0 < \alpha \le \log \frac{100}{99},$$

the polynomial $T_{\alpha,3}[(z-1)^3]$ has two nonreal zeros. Consequently $T_{\alpha,3}$ is not zero-sector reducing, and Question 6.1 of [1] has a negative answer.

Proof. Put $q = e^{-\alpha}$, so $99/100 \le q < 1$. The image of the polynomial with a triple positive root is

$$P_q(z) = q^{27}z^3 - 3q^8z^2 + 3qz - 1.$$

For $az^3 + bz^2 + cz + d$, the discriminant is

$$b^2c^2 - 4ac^3 - 4b^3d - 27a^2d^2 + 18abcd.$$

Substitution and factorization give the exact identity

$$\mathrm{Disc}(P_q) = -27q^{18}(1-q^6)^2(q^{24} + 2q^{18} + 3q^{12} - 2q^6 - 3). \tag{3}$$

For completeness, putting $u = q^6$ reduces the factorization to

$$3 - 4u - 4u^2 + 6u^3 - u^6 = -(u-1)^2(u^4 + 2u^3 + 3u^2 - 2u - 3).$$

Bernoulli's inequality gives $q^r \ge 1 - r/100$ for each positive integer r, while $q^6 \le 1$. Therefore

$$\begin{aligned} q^{24} + 2q^{18} + 3q^{12} - 2q^6 - 3 &\ge \left(1 - \frac{24}{100}\right) + 2\left(1 - \frac{18}{100}\right) + 3\left(1 - \frac{12}{100}\right) - 2 - 3 \\ &= \frac{1}{25} > 0. \end{aligned}$$

Since $0 < q < 1$, equation (3) is strictly negative. A real cubic with negative discriminant has one real zero and one nonreal conjugate pair.

Let ζ be one of the nonreal zeros. The input $(z-1)^3$ has all its zeros in $S(\theta)$ for every $\theta > 0$. Choose

$$0 < \theta < \min\{|\arg \zeta|, \pi/2\}.$$

Then $\zeta \notin S(\theta)$, and hence it cannot belong to any $S(\gamma)$ with $0 \le \gamma < \theta$. This contradicts (1). The argument uses positive half-angles and is independent of the convention at $\theta = 0$. □

Remark 2.2 (A rational certificate). At the particular value $\alpha = \log(100/99)$, every coefficient of P_q is rational. The proof consists of a rational polynomial identity and an explicit rational strict inequality. No numerical root approximation or finite-height exclusion enters the argument.

2.2 A quantitative third-difference obstruction

Let $g : \mathbb{N}_0 \to \mathbb{R}$ be any real sequence, and set

$$T_t^g[z^k] = e^{-tg(k)}z^k, \qquad t \ge 0. \tag{4}$$

For a fixed $m \ge 0$, normalize its four relevant values by

$$c_m = \min_{0 \le i \le 3} g(m+i), \qquad h_i = g(m+i) - c_m, \qquad M = \max_{0 \le i \le 3} h_i.$$

Thus $0 \le h_i \le M$. Subtracting c_m multiplies the cubic factor of the transformed polynomial by a positive scalar and does not change its zeros. Its third difference is unchanged:

$$d = \Delta^3 g(m) = h_3 - 3h_2 + 3h_1 - h_0.$$

Lemma 2.3 (Quantitative cubic obstruction)**.** *If $d \neq 0$, then $M > 0$, and $T_t^g[z^m(z-1)^3]$ has two nonreal zeros for every*

$$0 < t < \frac{d^2}{192M^3}. \tag{5}$$

More precisely, the normalized cubic

$$P_t(z) = e^{-th_3}z^3 - 3e^{-th_2}z^2 + 3e^{-th_1}z - e^{-th_0}$$

satisfies, for every $t \geq 0$,

$$\operatorname{Disc}(P_t) \leq -27d^2t^2 + 5184M^3t^3. \tag{6}$$

Proof. The exact discriminant is

$$\begin{aligned} D(t) = {} & 81e^{-t(2h_1+2h_2)} - 108e^{-t(3h_1+h_3)} - 108e^{-t(h_0+3h_2)} \\ & - 27e^{-t(2h_0+2h_3)} + 162e^{-t(h_0+h_1+h_2+h_3)}. \end{aligned} \tag{7}$$

Expanding through second order gives the polynomial identities

$$D(0) = D'(0) = 0, \qquad D''(0) = -54d^2.$$

For instance, the second identity of nonzero degree is equivalent to

$$\begin{aligned} & 81(2h_1+2h_2)^2 - 108(3h_1+h_3)^2 - 108(h_0+3h_2)^2 \\ & \quad - 27(2h_0+2h_3)^2 + 162(h_0+h_1+h_2+h_3)^2 = -54d^2. \end{aligned}$$

Every exponential rate in (7) belongs to $[0, 4M]$, and the sum of the absolute numerical weights is $81 + 108 + 108 + 27 + 162 = 486$. It follows that

$$|D^{(3)}(s)| \leq 486(4M)^3 = 31104M^3 \qquad (s \geq 0).$$

Taylor's theorem with its third-order remainder therefore gives

$$\left|D(t) + 27d^2t^2\right| \leq 5184M^3t^3.$$

This proves (6); the right side is strictly negative under (5). Finally,

$$T_t^g[z^m(z-1)^3] = e^{-tc_m}z^m P_t(z),$$

so the nonreal cubic zeros are zeros of the original transformed polynomial. Every constant in the bound is explicit. □

Corollary 2.4 (Every superquadratic power fails near zero)**.** *For every real $\beta > 2$, let*

$$d_\beta = 3^\beta - 3 \cdot 2^\beta + 3.$$

Then $d_\beta > 0$, and $T_{\alpha,\beta}[(z-1)^3]$ has two nonreal zeros whenever

$$0 < \alpha < \frac{d_\beta^2}{192\,3^{3\beta}}.$$

Proof. The repeated finite-difference formula gives

$$d_\beta = \beta(\beta-1)(\beta-2)\int_{[0,1]^3}(u+v+w)^{\beta-3}\,du\,dv\,dw > 0.$$

For $2 < \beta < 3$, first apply the formula to $(\varepsilon + x)^\beta$ and let $\varepsilon \downarrow 0$. The integrand is locally integrable at the origin; monotone convergence proves the identity. For $\beta \geq 3$, the formula applies directly. The lemma with $m = 0$, $g(k) = k^\beta$, and $M = 3^\beta$ now proves the assertion. □

Proposition 2.5 (The size of the emerging imaginary parts)**.** *Under the hypotheses of Lemma 2.3, the two nonreal zeros have imaginary parts of magnitude*

$$\frac{\sqrt{3}}{2}|d|^{1/3}t^{1/3} + O(t^{2/3}) \qquad (t \downarrow 0).$$

Their arguments have the same leading magnitude.

Proof. Put $t = u^3$ and $z = 1 + uw$. The expression

$$Q_u(w) = u^{-3}P_{u^3}(1 + uw)$$

extends analytically to $u = 0$, because $P_0(z) = (z-1)^3$. Its value there is $Q_0(w) = w^3 - d$. Since $d \neq 0$, these three roots are simple, so the analytic implicit-function theorem gives $w_j(u) = \omega_j + O(u)$, where $\omega_j^3 = d$. Thus $z_j(t) = 1 + t^{1/3}\omega_j + O(t^{2/3})$. The two nonreal cube roots of the nonzero real number d have imaginary magnitude $(\sqrt{3}/2)|d|^{1/3}$. Finally, $\arg(1+\eta) = \operatorname{Im}\eta + O(|\eta|^2)$ near $\eta = 0$. □

The exact rational example above is stronger than the conservative general time bound in this particular case. Neither result asserts failure for every positive time when $\beta > 2$. The assertion is failure for all sufficiently small positive times, plus the separately certified value $\alpha = \log(100/99)$ at $\beta = 3$.

2.3 Rigidity of diagonal semigroups

Theorem 2.6 (Quadratic rigidity)**.** *Let $g : \mathbb{N}_0 \to \mathbb{R}$. The following are equivalent.*

1. *For every $t \geq 0$, T_t^g preserves polynomials whose zeros all lie in $[0, \infty)$.*
2. *There exists a sequence $t_j > 0$, $t_j \to 0$, such that each $T_{t_j}^g$ preserves polynomials whose zeros all lie in $[0, \infty)$.*
3. *There are real constants A, B, C, with $A \geq 0$, such that*

$$g(k) = Ak^2 + Bk + C \qquad (k \in \mathbb{N}_0).$$

Furthermore, T_t^g is zero-sector reducing for every $t > 0$ if and only if $g(k) = Ak^2 + Bk + C$ with $A > 0$.

Proof. The first assertion implies the second. Suppose the second holds. If $\Delta^3 g(m) \neq 0$ for some m, then Lemma 2.3 produces nonreal zeros for $T_{t_j}^g[z^m(z-1)^3]$ whenever j is large enough. This is impossible. Thus $\Delta^3 g(m) = 0$ for every $m \geq 0$, and successive summation of finite differences gives

$$g(k) = g(0) + k\Delta g(0) + \binom{k}{2}\Delta^2 g(0).$$

This is a quadratic polynomial $Ak^2 + Bk + C$ on $\mathbb{N}_0$.

The image of $(z-1)^2$ has discriminant

$$4e^{-2tg(1)}\left(1 - e^{-t\Delta^2 g(0)}\right) = 4e^{-2tg(1)}(1 - e^{-2At}).$$

Nonnegativity for even one $t_j > 0$ forces $A \geq 0$.

Conversely, the multiplier e^{-Atk^2} preserves nonnegative real-rootedness for $A \geq 0$; this is the classical Gaussian multiplier property, also supplied by [1, Theorem 2.5] at $\theta = 0$. Multiplication

by e^{-tC} and substitution $z \mapsto e^{-tB}z$ preserve nonnegative real-rootedness, which proves the first assertion.

If every T_t^g, $t > 0$, is zero-sector reducing, it also preserves $S(0)$. Indeed, apply sector reduction to every arbitrarily small positive θ and use $\bigcap_{\theta>0} S(\theta) = S(0)$. The established equivalence therefore applies. If $A = 0$, the operator is merely a nonzero scalar and a positive dilation; the angles of the roots of $z^2 - 2(\cos\theta)z + 1$ remain $\pm\theta$, so strict sector reduction fails.

If $A > 0$, [1, Theorem 2.5] supplies the uniform half-angle

$$\gamma(t,\theta) = \arccos\bigl(\min\{1, e^{At}\cos\theta\}\bigr) < \theta, \qquad 0 < \theta < \pi/2.$$

The additional positive dilation and scalar do not alter root arguments. This proves strict sector reduction for every $t > 0$. □

Remark 2.7 (The classification is restricted to semigroups). Theorem 2.6 classifies the stated exponential families under the common small-time preservation hypothesis. It is not a characterization of all individual zero-sector reducing operators, and therefore is not an answer to the broader Question 6.2 of [1].

Methodological provenance. Repeated-root tests and discriminant geometry have a substantial history in multiplier-sequence theory; see, for example, Passare–Rojas–Shapiro [2]. We do not claim that using a discriminant is a new principle. The actual calculations supplied here are the rational sign certificate, the explicit square $-27(\Delta^3 g(m))^2$ in the Taylor coefficient, its uniform remainder bound, and the consequences with the quantifiers displayed above. Their historical priority remains unverified.

3 A printed Wolstenholme criterion and its novelty obstruction

For a prime p, put

$$C_p = \binom{2p-1}{p-1}, \qquad R_j = R_j(p) = \sum_{a=1}^{p-1} a^{-j}.$$

Rational congruences modulo p^r mean that the difference has p-adic valuation at least r. When all denominators are prime to p, this is the usual congruence in $\mathbb{Z}_{(p)}$.

Romeo Meštrović, *Congruences for Wolstenholme primes*, Czechoslovak Mathematical Journal **65** (2015), 237–253, Remark 1.1 on p. 240 [3], poses the converse question whether a prime satisfying

$$C_p \equiv 1 + 2pR_1 + \frac{2}{3}p^3R_3 \pmod{p^7} \tag{8}$$

must be a Wolstenholme prime. The forward implication for Wolstenholme primes is Corollary 1.1 of the same paper. The conjecture also occurs as Remark 1 in arXiv:1108.4178v1 (2011).

The exact conclusion. For every prime p, (8) holds if and only if p is a Wolstenholme prime, where the latter means $C_p \equiv 1 \pmod{p^4}$. In fact, for every prime $p \geq 11$,

$$C_p - 1 - 2pR_1 - \frac{2}{3}p^3R_3 \equiv 2p^2R_1^2 \pmod{p^7}. \tag{9}$$

Elementary harmonic valuations. If $1 \leq j < p-1$, then $R_j \equiv 0 \pmod{p}$: inversion permutes $\mathbb{F}_p^\times$, and the sum of jth powers over that group is zero. For odd j with $j+1 < p-1$, reflection gives

$$R_j = \sum_{a=1}^{p-1}(p-a)^{-j} \equiv -R_j - jpR_{j+1} \pmod{p^2}.$$

Thus $R_j \equiv 0 \pmod{p^2}$ for these odd j. In particular, for $p \geq 11$,

$$v_p(R_1), v_p(R_3), v_p(R_5) \geq 2.$$

Proof of the square defect. Define

$$F_p(z) = \prod_{a=1}^{p-1}(1 + z/a).$$

For odd p, $F_p(p) = C_p$ and $F_p(-p) = \prod_{a=1}^{p-1}(a-p)/a = 1$ exactly. The p-adic logarithm, which is convergent for every factor under consideration, therefore yields

$$\log C_p = \log F_p(p) - \log F_p(-p) = 2 \sum_{\substack{j \geq 1 \\ j \text{ odd}}} \frac{p^j}{j} R_j. \tag{10}$$

For $p \geq 11$, the $j = 5$ summand has valuation at least 7. For $j \geq 7$, $j - v_p(j) \geq 7$, so every remaining term belongs to $p^7\mathbb{Z}_p$. Indeed, the only borderline value $j = 7$ has $v_p(7) = 0$, and for $j \geq 8$ a positive $v_p(j)$ implies $j \geq 11$ and $v_p(j) \leq j - 7$. Consequently

$$\log C_p \equiv L := 2pR_1 + \frac{2}{3}p^3 R_3 \pmod{p^7}.$$

Here $v_p(L) \geq 3$. In the exponential series all terms of degree $r \geq 3$ have valuation at least $3r - v_p(r!) \geq 7$ for $p \geq 11$. For example, $v_p(r!) \leq (r-1)/(p-1)$ gives $3r - v_p(r!) \geq 3r - (r-1)/10 > 7$ for $r \geq 3$. Thus

$$C_p \equiv 1 + L + \frac{1}{2}L^2 \pmod{p^7}.$$

The product of $2pR_1$ and $(2/3)p^3R_3$ has valuation at least 8, and the square of the latter has valuation at least 10. This proves (9).

Since $v_p(R_1) \geq 2$,

$$2p^2R_1^2 \equiv 0 \pmod{p^7} \quad \Longleftrightarrow \quad v_p(R_1) \geq 3.$$

The same logarithmic expansion reduced modulo p^4 gives $C_p \equiv 1 + 2pR_1 \pmod{p^4}$, so this is precisely the Wolstenholme condition. This proves the result for $p \geq 11$.

For the remaining primes, the exact defects $\Delta_p = C_p - 1 - 2pR_1 - (2/3)p^3R_3$ are

p	C_p	Δ_p	$v_p(\Delta_p)$
2	3	$-22/3$	1
3	10	$-81/4$	4
5	126	15625/2592	6
7	1716	50706719/36000	6

None satisfies (8); none is Wolstenholme. The all-prime assertion follows.

Why this is not a G5 success. Meštrović, *On the mod p^7 determination of $\binom{2p-1}{p-1}$*, Rocky Mountain Journal of Mathematics **44** (2014), 633–648, DOI 10.1216/RMJ-2014-44-2-633, Theorem 1.1 [4] (arXiv:1108.1174v1), already proves

$$C_p \equiv 1 - 2pR_1 - 2p^2R_2 + 2p^2R_1^2 \pmod{p^7} \qquad (p \geq 11). \tag{11}$$

Reflection, without a Wolstenholme assumption, gives

$$2R_1 + pR_2 + p^2R_3 + p^3R_4 \equiv 0 \pmod{p^6}, \qquad 2R_3 + 3pR_4 \equiv 0 \pmod{p^4}.$$

To justify the first formula, expand $(p-a)^{-1}$ through degree 5; the omitted p^4R_5 and p^5R_6 terms belong to $p^6\mathbb{Z}_p$. For the second formula expand $(p-a)^{-3}$; the omitted $6p^2R_5$ and $10p^3R_6$ terms belong to $p^4\mathbb{Z}_p$. Eliminating R_4 gives

$$2R_1 + pR_2 + \tfrac{1}{3}p^2R_3 \equiv 0 \pmod{p^6}.$$

Substituting this in (11) immediately gives (9). The 2014 paper's Remark 1.6 already uses the same square $2p^2R_1^2$ to prove a neighboring characterization. The generic reflection relation also occurs in the proof of Corollary 1.1 of the 2015 paper. Therefore a new proof of the printed criterion cannot honestly be advertised as a new noncancellation or height estimate. An explicit later resolution was not located in the targeted search; that absence is not a proof of priority.

4 A realizability obstruction to transferring tower congruences

Fix an integer $k \geq 1$. We call an integer sequence $(a_n)_{n\geq 1}$ k-congruent if

$$a_{mp^r} \equiv a_{mp^{r-1}} \pmod{p^{kr}} \qquad (p \text{ prime},\ r \geq 1,\ m \geq 1,\ p \nmid m).$$

The following characterization is a direct Möbius-inversion argument, of the same type as classical Gauss–Dold congruences. It is included as a logical obstruction, not as a new solution of a named problem.

Proposition 4.1 (Möbius characterization)**.** *A sequence (a_n) is k-congruent if and only if there is an integer sequence (b_n) such that*

$$a_n = \sum_{d|n} d^k b_d \qquad (n \geq 1).$$

In that case $b_n = n^{-k}\sum_{d|n}\mu(n/d)a_d$, so the sequence (b_n) is unique.

Proof. If the divisor representation holds and $p \nmid m$, then

$$a_{mp^r} - a_{mp^{r-1}} = p^{kr}\sum_{d|m} d^k b_{dp^r},$$

which proves the congruence. Conversely, define $c_n = \sum_{d|n}\mu(n/d)a_d$. For $p^r \| n$, write $n = p^r m$ with $p \nmid m$. Terms with p-exponent less than $r-1$ have zero Möbius weight, and the remaining terms give

$$c_n = \sum_{d|m}\mu(m/d)(a_{dp^r} - a_{dp^{r-1}}).$$

Every summand is divisible by p^{kr}. This holds for every $p^r \| n$, so $n^k \mid c_n$. Set $b_n = c_n/n^k$ and apply Möbius inversion. □

Theorem 4.2 (Arbitrary exceptional-prime sets)**.** *Let E be any subset of the prime numbers. For every $k \geq 1$ there is a positive, multiplicative integer sequence (a_n), with $a_1 = 1$, such that*

1. *a_n is k-congruent;*
2. *for every prime p and every $r \geq 1$,*

$$v_p(a_{p^r} - a_{p^{r-1}}) = kr + \mathbf{1}_E(p);$$

3. *for all* $n \geq 1$, $a_n \leq n^{k+2}$.

Consequently, positivity, multiplicativity, polynomial growth, and uniform tower supercongruences do not, by themselves, imply that the primes with one additional digit of congruence precision are infinite, finite, nonempty, or of any prescribed density.

Proof. Put $c_p = p$ if $p \in E$ and $c_p = 1$ otherwise. Define a multiplicative integer sequence b by $b_1 = 1$ and $b_{p^r} = c_p$ for every $r \geq 1$. Set $a_n = \sum_{d|n} d^k b_d$. This is a positive multiplicative integer sequence, as the Dirichlet convolution of two multiplicative functions. The proposition proves all the tower congruences, and

$$a_{p^r} = 1 + c_p \sum_{j=1}^{r} p^{kj}, \qquad a_{p^r} - a_{p^{r-1}} = c_p p^{kr},$$

proves the exact valuation statement. For every d, $b_d \leq \mathrm{rad}(d) \leq d$; hence

$$a_n \leq \sum_{d|n} d^{k+1} \leq n^{k+1} \tau(n) \leq n^{k+2}.$$

The final logical conclusion follows because E was arbitrary. □

Remark 4.3 (Scope). The sequence constructed in this theorem depends on E and is not asserted to come from a fixed hypergeometric motive, a fixed Laurent polynomial, or the actual Fermat or Wolstenholme quotients. The theorem does not prove independence of any established arithmetic conjecture from a formal axiom system. Its exact conclusion is that an argument using only the four displayed properties cannot force a horizontal prime-distribution conclusion. A successful arithmetic transfer must use additional structure of its particular sequence.

5 What these results do and do not transfer to

5.1 Zero flows and the Riemann hypothesis

The real-zero question has a genuine conceptual connection to heat deformations. In the standard de Bruijn–Newman framework, the Riemann hypothesis is equivalent to $\Lambda \leq 0$. Rodgers and Tao proved $\Lambda \geq 0$ [6]. The obstruction in Section 2 concerns arbitrary diagonal semigroups acting on real polynomials. It provides no upper bound on Λ, no uniform estimate for the specific theta kernel defining the Riemann Ξ-function, and no proof that its zeros are real. Its useful implication for research design is precise: replacing a quadratic multiplier exponent by a superquadratic exponent cannot be justified as a universal small-time improvement in sector reduction.

5.2 Exceptional primes and the direction of the abc implication

For base 2, a Wieferich prime is an odd prime satisfying $2^{p-1} \equiv 1 \pmod{p^2}$. A Wolstenholme prime satisfies $C_p \equiv 1 \pmod{p^4}$. The criterion proved above is a statement about each prime individually. It does not prove that primes satisfying either exceptional condition occur infinitely often. The construction in the preceding section isolates the logical gap between precision at all powers of a fixed prime and distribution across different primes.

Silverman's theorem obtains infinitude of *non-Wieferich* primes from the abc conjecture [7]. The negation in that conclusion matters: it is not a theorem asserting infinitely many Wieferich primes. No argument here proves abc, supplies the required height/radical bound, or removes the hypothesis of that conditional implication. Similarly, proving the square-defect criterion does not produce any new prime with $v_p(R_1) \geq 3$.

5.3 BSD

The Birch and Swinnerton-Dyer conjecture concerns the agreement of the arithmetic rank of an elliptic curve with the order of vanishing of its L-function at $s = 1$, together with a leading-coefficient formula in its full form; see the official problem description [8]. No rank comparison, leading-coefficient formula, or finiteness theorem for a Tate–Shafarevich group follows from the polynomial and harmonic calculations in this draft. A transfer to BSD would require such additional arithmetic input, which is absent.

6 Corpus provenance and the six-gate audit

6.1 What was inspected

The source inventory was pinned to the repository *Verifiable Mathematical Reasoning Corpus* [9], commit

`c30d40119134a59eda6fa5cc6331f27c8e67d034`.

The commit was dated 6 November 2026, 16:24:52 UTC. The live inventory contained 130 `.tex` files. Of these, 128 matched an existing local snapshot by Git blob SHA-1, and the two additional files, numbered 129 and 130, were retrieved separately. This checks the identity of the working corpus; it is not a verification of all its mathematical claims.

The reading was targeted at mechanisms relevant to the requested problems, including the following documents and themes.

Corpus item	Theme examined	Role in this draft
013	Profinite recognition, Mahler equations, Witt-type escape	Distinguish finite certificates from a theorem controlling every remaining obstruction.
043	Local certificates, stability, and concavity	Compare the local valuation method and check that the power-sum target in Appendix A is not the different block-reduction assertion discussed there.
053	Restriction classes and Baker's $L(1, \chi)$ question	Identify the need for a genuine nonvanishing term, rather than a formal equivalence alone.
112	Period obstructions for weighted binary expansions	Separate a prime's lifting valuation from an infinitude claim over primes.
126	Real collisions in the Gaussian heat flow	Motivate an adversarial preservation test at a repeated real zero.
129–130	Certified prime lifting; sparse modular squaring	Use exact finite arithmetic as a certificate without extrapolating it to an infinite distribution theorem.

No unverified theorem from the corpus is used as a premise of the proofs above. The structural synthesis is methodological: retain an exact defect, prove that its first surviving term cannot cancel, and distinguish the resulting local assertion from a global prime or zero-distribution statement. This is a description of the reasoning workflow, not a claim to have invented a historically new general principle.

6.2 Gate-by-gate judgment

The research brief requires all six gates simultaneously. The following judgment is deliberately stricter than merely checking a proof for algebraic correctness.

Gate	Requirement	Judgment
G1	Pre-existing named target	The cited numbered questions and conjectures predate this draft. The auxiliary semigroup classification and local valuation refinements are not counted as additional named-problem solutions.
G2	Original statement unchanged	The Cardon operator, its parameters, and its sectors are retained. The harmonic and power-sum congruences retain the original constants and moduli. Conjecture 11 is refuted only in its first assertion.
G3	Actually still unresolved	The original publications label the targets as open questions or conjectures. This does not certify their status in November 2026. For the Wolstenholme criterion, earlier established arithmetic already yields the result by an elementary substitution.
G4	Unconditional complete proof	The displayed counterexamples, the general cubic obstruction, the stated semigroup classification, the harmonic equivalence, and the power-sum theorem have proofs covering all their quantifiers. No numerical experiment is used in place of a universal proof.
G5	Substantive new mechanism	An actual negative discriminant bound and exact local valuation are proved. However, their historical methodological novelty is not established. Repeated-root discriminant tests are classical, and the Wolstenholme conclusion is a consequence of known congruences. No G5 pass is asserted.
G6	No earlier duplicate resolution	Targeted searches inspected the original papers, later indexed references, and related multiplier-sequence literature. No exhaustive nonduplication certificate is available. Search absence is not used to assert priority.

Overall status. There are complete mathematical answers to the printed assertions specified in this draft. There are *zero results certified here to satisfy all of G1–G6.* In particular, this draft must not be cited as a verified historically first solution of a major open problem or as a proof of any of the five headline conjectures in the research brief.

6.3 Checks performed and their limits

The discriminant factorization and its general second-order Taylor coefficient were checked symbolically. The rational sign certificate was checked using exact arithmetic. Additional modular checks covered every prime $11 \le p < 1000$ and $p = 16843$, including a Wolstenholme case; the four smaller primes were evaluated as exact fractions. The telescoping identity and its valuation refinement were checked over finite grids. These checks supplement the proofs and help detect transcription errors. They neither establish historical priority nor turn a statement about finitely many primes into an infinitude theorem.

A The unchanged power-sum Conjecture 19

Write $S_j(n) = \sum_{a=1}^{n-1} a^j$, including $S_0(n) = n - 1$. Meštrović, *On a Congruence Modulo n^3 Involving Two Consecutive Sums of Powers*, Journal of Integer Sequences **17** (2014), Article

14.8.4, Conjecture 19 [5], states that

$$\sum_{i=0}^{k}(1+n(k+1-i))\binom{2k+2}{2i}S_{2i}(n)\equiv -1 \pmod{n^3} \tag{12}$$

for every $k\geq 1$ and every even positive integer n. The preprint version arXiv:1211.4570v1 (2012) labels this Conjecture 2.17. The following is a complete elementary proof of the unchanged assertion, but no historical priority or G5 success is claimed.

Put $m=2k+2\geq 4$, and denote the sum on the left by $T_m(n)$. The binomial formula gives

$$\sum_{\substack{0\leq j\leq m-2\\ j \text{ even}}}\binom{m}{j}a^j=\frac{(a+1)^m+(a-1)^m}{2}-a^m,$$

and

$$\sum_{\substack{0\leq j\leq m-2\\ j \text{ even}}}(m-j)\binom{m}{j}a^j=\frac{m}{2}((a+1)^{m-1}-(a-1)^{m-1}).$$

Summation over $a=1,\dots,n-1$ telescopes, yielding the exact identity

$$4(T_m(n)+1)=(m+2)n^m-2(n-1)^m+mn(n-1)^{m-1}-mn+2. \tag{13}$$

Expanding the right side, the coefficients of n^0,n^1,n^2 vanish. For $3\leq j<m$, the coefficient of n^j is $(-1)^j(j-2)\binom{m}{j}$; the coefficient of n^m is $2m$. Hence

$$\frac{4(T_m(n)+1)}{n^3}=\sum_{j=3}^{m-1}(-1)^j(j-2)\binom{m}{j}n^{j-3}+2mn^{m-3}. \tag{14}$$

If $m=2r$, then

$$\binom{m}{3}=\frac{2r(2r-1)(r-1)}{3}\equiv 0 \pmod 4,$$

because $r(r-1)$ is even and division by the odd number 3 cannot remove a factor of 4. Thus the $j=3$ term of (14) is divisible by 4. The $j=4$ term, when present, is $2\binom{m}{4}n$, divisible by 4 for even n. Every $j\geq 5$ term is divisible by 4 because n^{j-3} is, and the final term is divisible by 4 since m is even. The right side of (14) is therefore divisible by 4. Consequently $(T_m(n)+1)/n^3$ is an integer, proving (12) for all the required k and n.

This argument closes the displayed assertion as mathematics. It uses an exact telescoping identity followed by elementary coefficient divisibility, precisely the kind of result excluded by the stated strict G5 requirement if a new noncancellation/height/zero estimate is required. Targeted web searches for Conjecture 19 and Conjecture 2.17 did not locate a later explicit proof; this limited negative finding is not evidence strong enough to assert G3 or G6.

An exact local valuation strengthening. The same expansion proves a more precise noncancellation statement. Let ℓ be a prime dividing n. If ℓ is odd, or if $\ell=2$ and $4\mid n$, then

$$v_\ell(T_m(n)+1)=3v_\ell(n)+v_\ell\binom{m}{3}-v_\ell(4). \tag{15}$$

This statement is proved here as mathematics; its historical novelty has not been established.

For $3\leq j<m$, the ratio of the absolute value of the degree-j coefficient in (14) to the degree-3 coefficient is

$$\frac{(j-2)\binom{m}{j}}{\binom{m}{3}}=\frac{6}{j(j-1)}\binom{m-3}{j-3}.$$

Put $a = v_\ell(n) \geq 1$. For every $j \geq 4$, the valuation of the jth term relative to the cubic term is at least

$$(j-3)a + v_\ell(6) - v_\ell(j(j-1)).$$

This is positive for odd ℓ: when $\ell = 3$ use $v_3(j(j-1)) \leq j-3$, and when $\ell \geq 5$ use $v_\ell(j(j-1)) \leq j-4$. For $\ell = 2$ and $a \geq 2$, it is at least $2(j-3)+1-(j-2) = j-3 \geq 1$. These elementary bounds follow because at most one of the two consecutive integers $j, j-1$ is divisible by ℓ, with the cases $j = 4, 5$ checked directly and the remaining cases bounded by $\log_\ell j$.

The final term in (14) has ratio

$$\frac{2m}{\binom{m}{3}} n^{m-3} = \frac{12}{(m-1)(m-2)} n^{m-3}.$$

For $m = 4$ this is $2n$, of positive valuation for every $\ell \mid n$. For even $m \geq 6$, at most one of $m-1, m-2$ is divisible by ℓ, and $v_\ell((m-1)(m-2)) \leq \lfloor \log_2(m-1) \rfloor \leq m-4$. The final term therefore also has strictly larger valuation than the cubic term. The latter is the unique term of least valuation, proving (15).

In particular, for $4 \mid n$,

$$v_2(T_m(n)+1) = 3v_2(n) + v_2(m) + v_2(m-2) - 3.$$

Thus the bound modulo n^3 is sharp in its 2-part whenever $m \equiv 4$ or $6 \pmod 8$ and $4 \mid n$.

B A counterexample to the first part of Conjecture 11

The same 2014 paper, Conjecture 11 [5] (arXiv:1211.4570v1, Conjecture 2.9), asserts that each of

$$S_n(n) \equiv \frac{n^3}{2} B_{n-1} \pmod{n^5}, \qquad S_{n-1}(n) \equiv n B_{n-1} \pmod{n^4}$$

is satisfied by no integer $n \geq 2$. Here $B_4 = -1/30$, and at $n = 5$ the first asserted nonexistence fails:

$$S_5(5) = 1 + 32 + 243 + 1024 = 1300, \qquad \frac{5^3}{2} B_4 = -\frac{25}{12}, \qquad 1300 + \frac{25}{12} = \frac{5^6}{12}.$$

The denominator 12 is a unit modulo 5^5, so this is an exact counterexample to the first universal claim, with the original statement and constants unchanged.

In fact the first congruence holds for every prime $p \geq 5$. For $p \geq 7$, Faulhaber's formula writes

$$S_p(p) = \sum_{j=1}^{p+1} \frac{1}{j} \binom{p}{j-1} p^j B_{p+1-j}.$$

The $j = 2$ term is $p^3 B_{p-1}/2$. The $j = 1$ term is zero. The $j = 3$ term is also zero because $B_{p-2} = 0$. For $4 \leq j \leq p-1$ with a nonzero Bernoulli factor, j is even; $\binom{p}{j-1}$ is divisible by p, while j and the denominator of B_{p+1-j} are prime to p; the latter assertion follows from the von Staudt–Clausen denominator theorem, since the nonzero even index lies strictly between 0 and $p-1$. These terms are divisible by p^5. The $j = p$ term involves B_1 and has valuation $p \geq 7$; the final $j = p+1$ term has valuation $p+1$. This proves the assertion for $p \geq 7$, and the displayed calculation proves the remaining $p = 5$.

This is a correction to a printed statement, not a claim that a substantial previously open problem has been solved. The mechanism is the classical Faulhaber formula; it does not meet the user's G5 novelty condition. The current prior-art search is insufficient to assert that this printed error has never previously been noticed.

C Reproducible exact checks

Compile this source twice with `pdflatex`; no shell escape, external bibliography, images, or included source files are needed. The following optional verification program uses Python 3 and SymPy. Save the listing as `verify.py` and run `python3 verify.py`. It performs symbolic identity checks and exact integer/rational diagnostics; it is not needed for the mathematical proofs.

```
from fractions import Fraction as F
from math import comb, isqrt
from pathlib import Path
import json
import sympy as s

q = s.symbols('q')
z = s.symbols('z')
P = q**27*z**3 - 3*q**8*z**2 + 3*q*z - 1
disc = s.discriminant(P, z)
factor = -27*q**18*(1-q**6)**2*(q**24+2*q**18+3*q**12-2*q**6-3)
assert s.expand(disc-factor) == 0
q0 = s.Rational(99, 100)
bracket = q0**24+2*q0**18+3*q0**12-2*q0**6-3
assert bracket >= s.Rational(1, 25)
assert factor.subs(q, q0) < 0

h0, h1, h2, h3 = s.symbols('h0 h1 h2 h3')
weights = [81, -108, -108, -27, 162]
rates = [2*h1+2*h2, 3*h1+h3, h0+3*h2,
         2*h0+2*h3, h0+h1+h2+h3]
delta = h3-3*h2+3*h1-h0
assert sum(weights) == 0
assert s.expand(sum(c*r for c,r in zip(weights,rates))) == 0
assert s.expand(sum(c*r*r for c,r in zip(weights,rates))/2+27*delta**2) == 0
assert sum(map(abs, weights))*4**3//6 == 5184

def prime(n):
    return n >= 2 and all(n % d for d in range(2,isqrt(n)+1))

def vp(n,p):
    if isinstance(n,F):
        return vp(n.numerator,p)-vp(n.denominator,p)
    if n == 0:
        return float('inf')
    v=0
    while n % p == 0:
        n//=p
        v+=1
    return v

prime_cases = [p for p in range(11,1000) if prime(p)]+[16843]
for p in prime_cases:
    mod=p**7
    inv=[pow(k,-1,mod) for k in range(1,p)]
    r1=sum(inv)%mod
    r3=sum(pow(a,3,mod) for a in inv)%mod
    C=comb(2*p-1,p-1)%mod
    residual=(C-1-2*p*r1-2*pow(3,-1,mod)*p**3*r3)%mod
    assert (residual-2*p*p*r1*r1)%mod == 0
    assert (residual == 0) == (C%p**4 == 1)
    if p == 16843:
        assert residual == 0
```

```
small_expected={2:F(-22,3),3:F(-81,4),5:F(15625,2592),7:F(50706719,36000)}
small_records=[]
for p,expected in small_expected.items():
    r1=sum((F(1,k) for k in range(1,p)),F(0))
    r3=sum((F(1,k**3) for k in range(1,p)),F(0))
    defect=comb(2*p-1,p-1)-1-2*p*r1-F(2,3)*p**3*r3
    assert defect == expected
    assert vp(defect,p)<7
    assert (comb(2*p-1,p-1)-1)%p**4 != 0
    small_records.append({'p':p,'defect':str(defect),'valuation':vp(defect,p)})

telescoping_cases=0
valuation_cases=0
for m in range(4,34,2):
    for n in range(2,42):
        T=sum((1+n*(m-j)//2)*comb(m,j)*sum(a**j for a in range(1,n))
              for j in range(0,m,2))
        rhs=(m+2)*n**m-2*(n-1)**m+m*n*(n-1)**(m-1)-m*n+2
        assert 4*(T+1) == rhs
        telescoping_cases+=1
        if n%2 == 0:
            assert (T+1)%n**3 == 0
        for ell in range(2,n+1):
            if prime(ell) and n%ell == 0 and (ell%2 or n%4 == 0):
                assert vp(T+1,ell) == 3*vp(n,ell)+vp(comb(m,3),ell)-vp(4,ell)
                valuation_cases+=1

assert F(sum(a**5 for a in range(1,5)))-F(5**3,2)*F(-1,30) == F(5**6,12)
report={
 'symbolic_discriminant_identity':True,
 'symbolic_general_Taylor_coefficients':True,
 'exact_q_99_100_certificate':True,
 'harmonic_prime_cases':len(prime_cases),
 'included_Wolstenholme_prime':16843,
 'small_prime_cases':small_records,
 'exact_telescoping_cases':telescoping_cases,
 'exact_local_valuation_cases':valuation_cases,
 'printed_nonexistence_counterexample':5,
 'scope':'Diagnostics supplement the universal proofs; no finite test establishes historical
    novelty or prime infinitude.'}
Path('verification_results.json').write_text(json.dumps(report,indent=2)+'\n')
print(json.dumps(report,indent=2))
```

The verified run reported 165 harmonic prime cases, 600 telescoping cases, and 765 local valuation cases, together with the four exact small-prime defects. A failed assertion indicates a reproducibility or transcription problem that should be investigated before using the corresponding formula.

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